SET 16: CUSTOMER DEMOGRAPHICS ANALYSIS

Aim

To analyze customer demographic information using R programming by creating a bar chart, pie chart, table, and interactive dashboard.

Dataset

The dataset contains Customer ID, Age, Gender, and Income.

R Code – Create Dataset

```
customer <- data.frame(
  Customer_ID = c(1, 2, 3),
  Age = c(28, 35, 42),
  Gender = c("Female", "Male", "Female"),
  Income = c(50000, 60000, 75000)
)

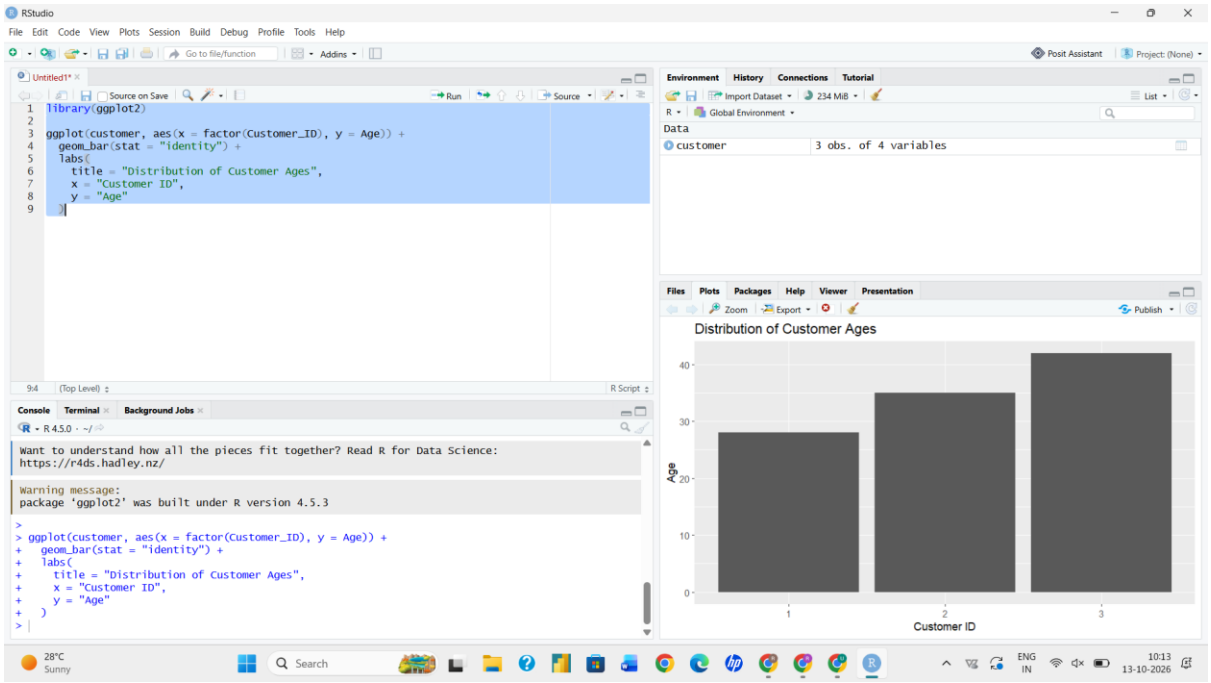
customer
```

Question 1: Create a Bar Chart to Visualize Customer Ages

R Code

```
library(ggplot2)

ggplot(customer, aes(x = factor(Customer_ID), y = Age)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Distribution of Customer Ages",
    x = "Customer ID",
    y = "Age"
  )
```



Result

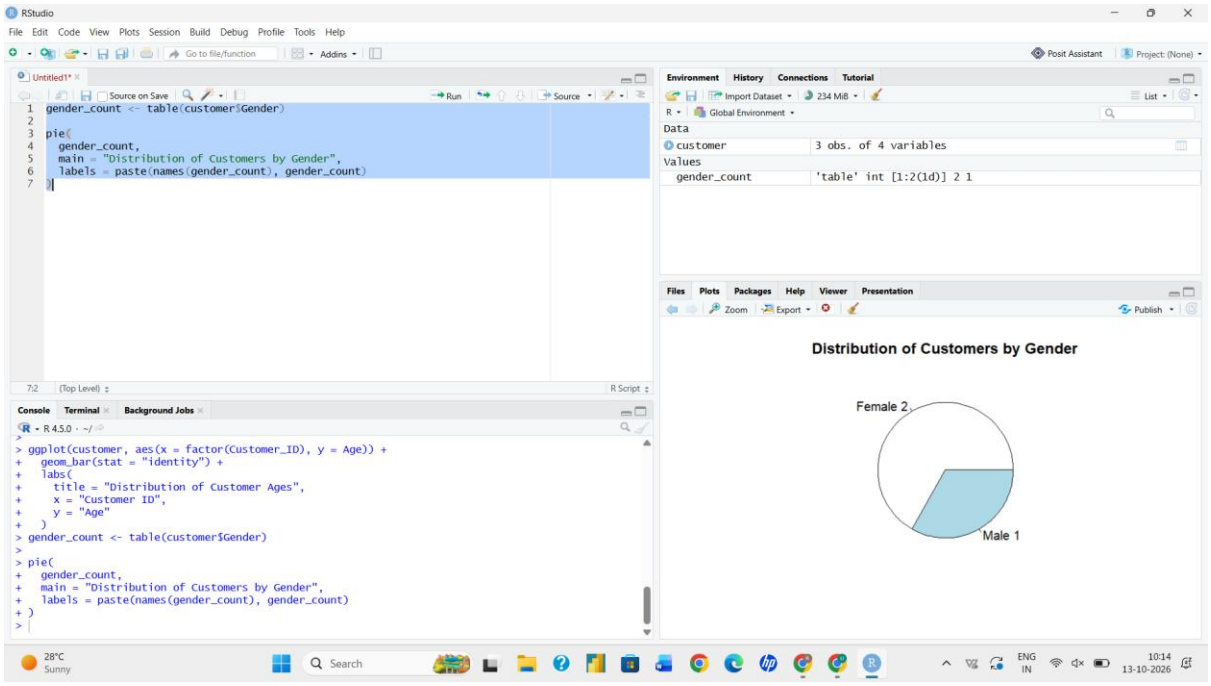
The bar chart displays the age of each customer. Customer 1 is 28 years old, Customer 2 is 35 years old, and Customer 3 is 42 years old.

Question 2: Generate a Pie Chart for Gender Distribution

R Code

```
gender_count <- table(customer$Gender)


pie(
  gender_count,
  main = "Distribution of Customers by Gender",
  labels = paste(names(gender_count), gender_count)
)
```



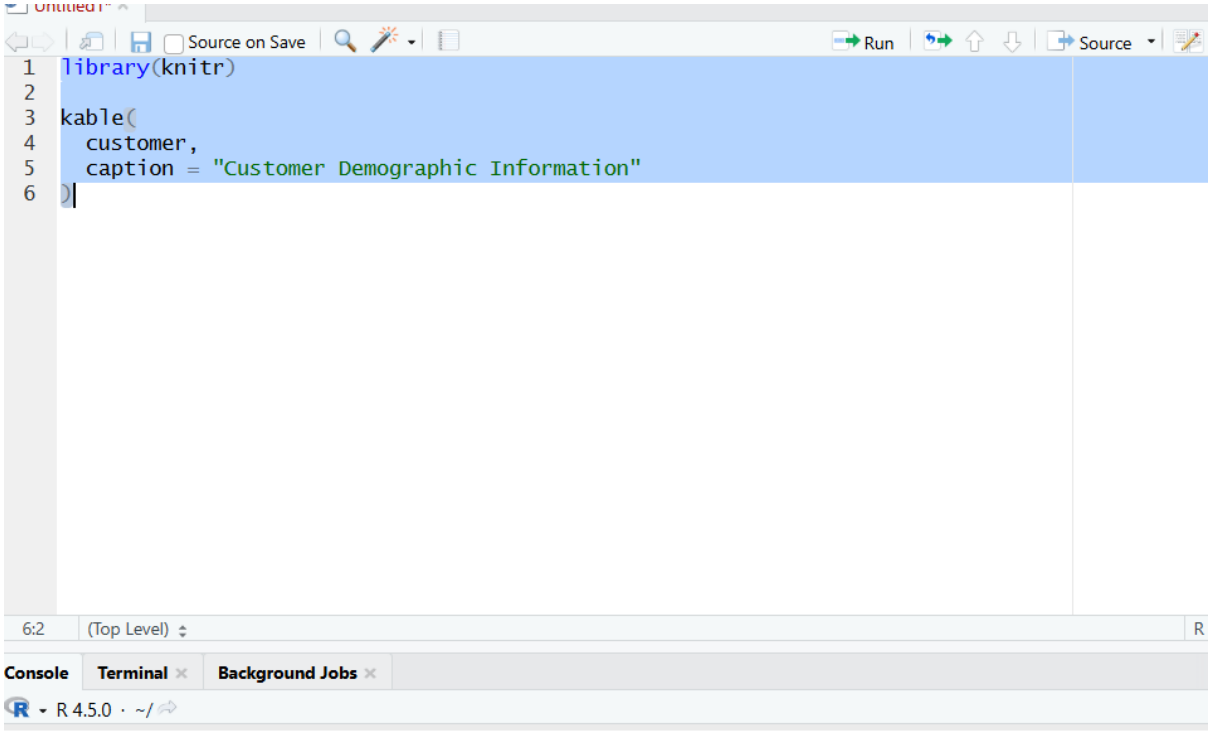
Result

The pie chart shows that there are 2 female customers and 1 male customer.

Question 3: Build a Table Showing Demographic Information

R Code

Customer



Result

The table displays Customer ID, Age, Gender, and Income for each customer.

Question 4: Create an Interactive Dashboard

R Code

library(shiny)

library(ggplot2)

library(plotly)

library(DT)

ui <- fluidPage(

```
  titlePanel("Customer Demographics Dashboard"),

  plotlyOutput("agePlot"),
  plotlyOutput("genderPlot"),
  DTOutput("customerTable")
)

server <- function(input, output) {

  output$agePlot <- renderPlotly({
    p <- ggplot(customer, aes(x = factor(Customer_ID), y = Age)) +
      geom_bar(stat = "identity") +
      labs(
        title = "Customer Ages",
        x = "Customer ID",
        y = "Age"
      )

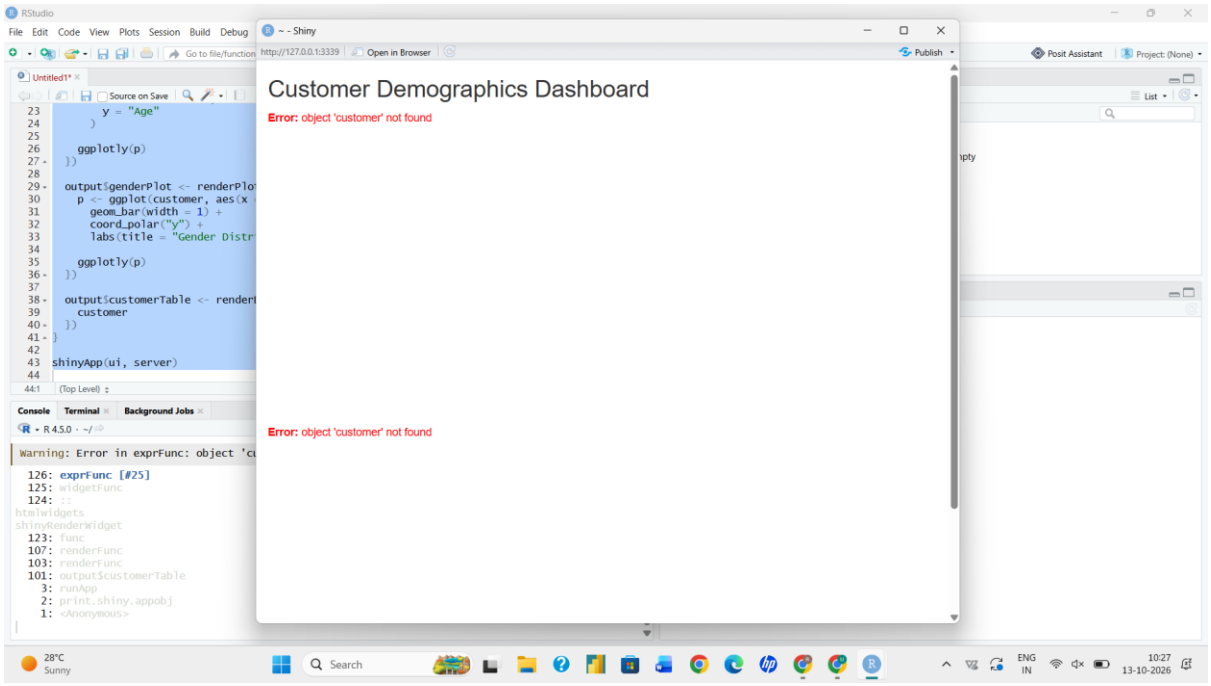
    ggplotly(p)
  })

  output$genderPlot <- renderPlotly({
    p <- ggplot(customer, aes(x = "", fill = Gender)) +
      geom_bar(width = 1) +
      coord_polar("y") +
      labs(title = "Gender Distribution")

    ggplotly(p)
  })
```

```
  output$customerTable <- renderDT({
    customer
  })
}

shinyApp(ui, server)
```



Result

An interactive dashboard is created containing the customer age chart, gender pie chart, and demographic table.

Conclusion

Customer demographic data was successfully analyzed using R. Charts, tables, and an interactive dashboard were created to understand the customer information.